

Amagi Architectural Class Definition

Institutional Anchor Document v1.1

DATE: 2026-05-12

DOCUMENT TYPE: Institutional Anchor

STATUS: Document v1.1

AUTHOR: Alexey Mikhailovich Burlai

ORCID: 0009-0001-4679-5967

LICENSE: Creative Commons Attribution–NonCommercial 4.0 International (CC BY-NC 4.0)

DOI: 10.5281/zenodo.20145198

REFERENCE CONTEXT: Amagi Framework Ecosystem

1. Scope and Purpose

This document establishes the formal definition of the Amagi architectural class and its irreducible architectural invariants.

It serves as the authoritative, non-negotiable reference for determining whether a system constitutes an authentic Amagi-class architecture.

This document exists to:

- define the Amagi architectural class as a distinct and bounded category;
- prevent mischaracterization, dilution, or structural imitation of the Amagi Framework;
- provide a stable institutional reference independent of any specific implementation;
- serve as public prior art and a governance anchor for certification, verification, and regulatory alignment.

This document defines an architectural class independently of jurisdiction, application domain, or regulatory regime.

This document does not disclose trade secrets, confidential implementation details, or domain-specific hard intellectual property.

2. Definition of the Amagi Architectural Class

Amagi is defined as an architectural class, not as an implementation pattern, product, feature set, software framework, or deployment model.

Membership in the Amagi architectural class is determined exclusively by satisfaction of the architectural invariants defined in this document.

Similarity in design intent, functional outcome, claimed purpose, or regulatory alignment does not constitute Amagi-class equivalence.

Amagi-class membership is a binary condition: systems either satisfy the defined invariants in full or do not belong to the Amagi architectural class.

Version 1.1 clarification: The binary condition applies to the invariant set as specified. A system may operate under reduced assurance mode (Section 4) with explicit architectural waiver, but such a system is not Amagi-class. It is a derivative architecture under commercial license (see AGIAM Technologies Founding Charter v1.1, DOI: 10.5281/zenodo.20145094).

3. Irreducible Architectural Invariants

An Amagi-class system shall satisfy all of the following invariants. These invariants are necessary and indivisible conditions for Amagi-class membership.

AMAGI-INVARIANT-001: Physical Non-Bypassability

Control enforcement shall be guaranteed by physical construction and hardware isolation.

Software convention, policy, runtime checks, virtualization, or best-effort enforcement mechanisms alone are insufficient.

AMAGI-INVARIANT-002: Irreducible Trusted Computing Base

A minimal Security Council Core (SCC) shall constitute the sole trusted control plane.

The SCC shall be structurally minimal, isolated, and amenable to formal verification.

AMAGI-INVARIANT-003: Hardware-Enforced Separation of Control and Cognition

Cognitive components shall be treated as untrusted and shall not possess direct authority over consequential actions, actuation paths, or state transitions.

Authority separation shall be enforced structurally and physically, not contractually or procedurally.

AMAGI-INVARIANT-004: Immutability of Critical Policy Structures

Safety-critical policy, access graphs, and enforcement logic shall be immutable post-deployment or fused into hardware or one-time-programmable memory.

Post-deployment modification, self-modification, or dynamic rewriting of these structures is not permitted.

AMAGI-INVARIANT-005: Complete Mediation

All consequential actions, actuation paths, and state transitions shall transit exclusively through the trusted control plane.

No side channels, bypass paths, or auxiliary control mechanisms may exist outside this mediation boundary.

AMAGI-INVARIANT-006: Complete Audit Integrity

Audit records shall be cryptographically protected, append-only, and structurally inaccessible to cognitive components.

Cognitive systems shall not be capable of modifying, suppressing, or selectively influencing audit visibility.

Invariant Boundary Condition

Removal, weakening, optionalization, substitution, or conditional application of any invariant defined above results in a system that is not Amagi-class.

Partial compliance is not defined.

4. Reduced Assurance Mode (NEW in v1.1)

4.1 Definition

A reduced assurance mode is an operational configuration in which one (and only one) of the six irreducible invariants is explicitly waived by architectural decision, documented risk assessment, and licensed commercial agreement.

4.2 Conditions for Reduced Assurance Mode

1. Explicit Waiver

The waived invariant must be explicitly identified in system documentation.

2. Risk Assessment

A documented risk assessment shall demonstrate that the waived invariant is compensated by alternative controls or that the residual risk is acceptable for the target deployment context.

3. License Requirement

Reduced assurance mode is permitted only under valid S-APL commercial license from AGIAM Technologies Ltd.

4. Prohibited Waivers

The following invariants shall not be waived under any circumstances:

- AMAGI-INVARIANT-001 (Physical Non-Bypassability)
- AMAGI-INVARIANT-005 (Complete Mediation)
- AMAGI-INVARIANT-006 (Complete Audit Integrity)

5. Permissible Waivers

Only the following invariants may be waived in reduced assurance mode:

- AMAGI-INVARIANT-002 (Irreducible TCB) — with SCC complexity increase formally verified
- AMAGI-INVARIANT-003 (Separation of Control and Cognition) — with additional software isolation layers
- AMAGI-INVARIANT-004 (Immutability) — with cryptographic update verification and rollback protection

4.3 Classification of Reduced Assurance Systems

A system operating in reduced assurance mode:

- Is not Amagi-class.
 - Is a Derivative Architecture under AGIAM Technologies Founding Charter v1.1 (DOI: 10.5281/zenodo.20145094).
 - Shall not claim Amagi-class certification or presumption of conformity.
 - May claim "Amagi-Derived" or "Amagi-Compatible" status with explicit waiver disclosure.
-

4.4 Relationship to Commercial Licensing

The AGIAM Technologies Founding Charter v1.1 defines Derivative Architecture criteria as:

- Core Principles: At least 5 of 6 (permissible under reduced assurance mode)
- Core Components: At least 3 of 6

This is not Amagi-class membership. It is a commercial deployment category with reduced structural guarantees.

5. Irreducibility Statement

The Amagi architectural class is irreducibly complex.

The defined invariants are not design preferences, implementation guidelines, or optimization targets.

They arise directly from the requirement to enforce safety, authority limitation, and compliance as structural properties, rather than procedural assurances or behavioral expectations.

Any simplification that removes, relaxes, externalizes, or defers enforcement of an invariant collapses the architecture into a weaker and categorically distinct class.

Version 1.1 clarification: Reduced assurance mode (Section 4) does not contradict irreducibility. It is an explicitly documented, licensed, and risk-assessed waiver for commercial contexts where full Amagi-class rigour is not required. The architecture itself remains irreducible; the waiver is a deployment-time configuration, not an architectural change.

6. Relationship to Open Publication

The Amagi Open Specification and related publications are released as public prior art to enable independent analysis and to prevent enclosure through patents or proprietary claims.

Open publication does not imply permissive implementation rights, waiver of architectural constraints, or automatic Amagi-class equivalence.

Claims of Amagi compatibility require demonstrable structural equivalence to the invariants defined in this document.

Version 1.1 addition: Claims of "Amagi-Derived" or "Amagi-Compatible" status require disclosure of any invariant waivers and valid S-APL license.

7. Relationship to Regulatory and Standards Frameworks

This document is intended to be compatible with, but not limited to, emerging regulatory and standards frameworks governing high-risk and safety-critical systems, including AI governance, critical infrastructure protection, and formally assured computing.

References to such frameworks are non-normative and do not constrain, modify, or supersede the architectural invariants defined herein.

8. Custodianship and Institutional Role

AGIAM acts as the architectural custodian of the Amagi architectural class.

Custodianship includes:

- maintenance of the canonical definition of Amagi-class invariants;
- definition of structural equivalence and verification criteria;
- preservation of architectural continuity across versions and implementations.

Custodianship does not imply ownership of intelligence, models, data, applications, or outcomes built within or around Amagi-class systems.

9. Relationship to Trade Secrets and Licensing

This document:

- does not grant access to confidential implementation knowledge;
- does not supersede licensing requirements for commercial deployment;
- does not disclose domain-specific optimization, calibration, verification environments, or manufacturing processes.

Implementation-level knowledge may remain protected under applicable trade secret, licensing, and contractual regimes.

10. Change Log

Version	Date	Changes	Author
1.0	2025-11-30	Initial release with 6 irreducible invariants	A.M. Burlai
1.1	2026-05-12	Added Section 4 (Reduced Assurance Mode); clarified binary membership vs commercial flexibility; added prohibited/permissible waiver matrix	A.M. Burlai

11. References

1. Burlai, A.M. (2025). Amagi Architectural Class Definition (IAA) v1.0. DOI: 10.5281/zenodo.17938573.
2. Burlai, A.M. (2026). AGIAM Technologies Founding Charter v1.1. DOI: 10.5281/zenodo.20145094.
3. Burlai, A.M. (2025). Amagi Open Specification v1.0. DOI: 10.5281/zenodo.17861024.

© 2026 AGIAM Technologies Ltd.